

A transmission line with $Z_o = 50 \Omega$ is terminated with an impedance $50 - j50 \Omega$.

Find the reflection coefficient, standing-wave ratio, and fraction of the incident power delivered to the load

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$$\text{SWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$P_L = P_{in}[1 - \rho^2]$$

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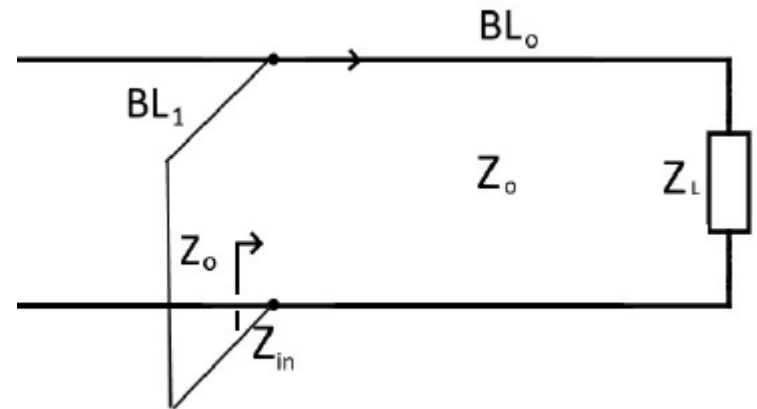
$$|\Gamma| = \rho$$

$$\Gamma = 0.2 - j0.4 = \frac{\sqrt{5}}{5} \angle -63.43^\circ$$

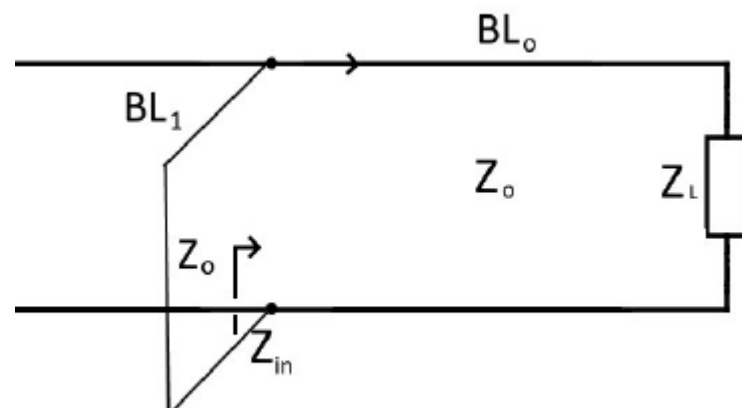
$$SWR = 2.618$$

$$\frac{P_L}{P_{in}} = 0.8 = 80\%$$

Given the following circuit and $\beta l_0 = \pi/4$, $\beta l_1 = \pi/8$, $Z_0 = 50\Omega$, and $Z_L = 75\Omega$, what is the input impedance?



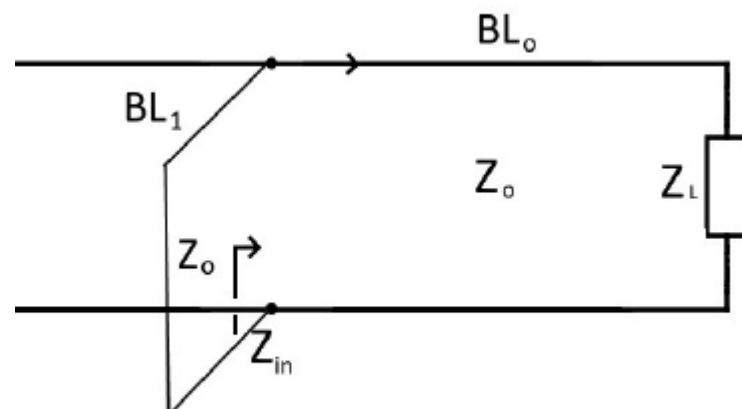
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- First solving for the impedance of the shorted section;

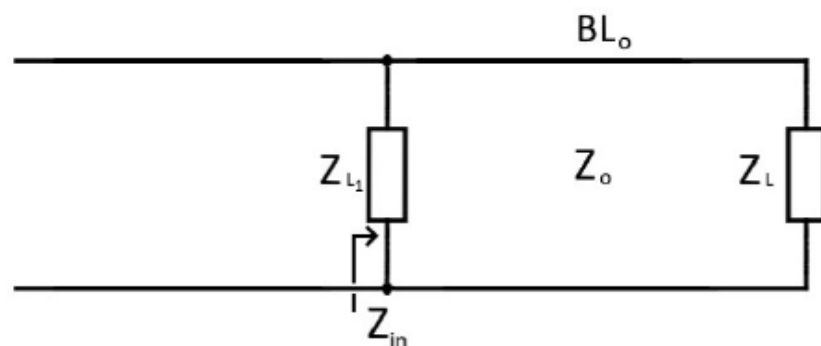
- $Z_1 = Z_0 \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)}$, where $Z_L = 0$
- $Z_1 = Z_0 \frac{(0) + jZ_0 \tan(\beta l)}{Z_0 + j(0) \tan(\beta l)} = jZ_0 \tan(\beta l)$
- $Z_1 = j(50) \tan(\pi/8)$
- $Z_1 = j20.71$

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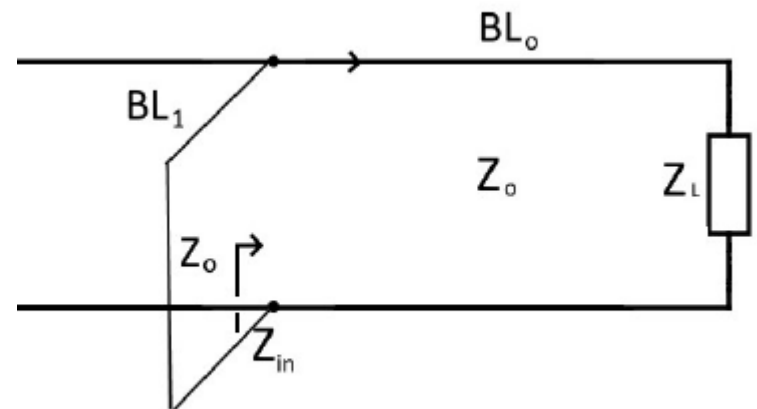


- Now solving for the impedance of the line to the load;

- $$Z_2 = Z_0 \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)}$$
- $$Z_2 = (50) \frac{(75) + j(50) \tan(\pi/4)}{(50) + j(75) \tan(\pi/4)}$$
- $$Z_2 = 46.154 - j19.231$$



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- Now combining the impedance from the load line and the shorted line;

- $$Z_{in} = \left(\frac{1}{Z_1} + \frac{1}{Z_2} \right)^{-1}$$

- $$Z_{in} = \left(\frac{1}{(j20.71)} + \frac{1}{(46.154 - j19.231)} \right)^{-1}$$

- $$Z_{in} = 9.283 + j20.413$$

